# Northwestern at AraGenre 2026 Definition-Guided Dense Retrieval Using BGE-M3 for Hierarchical Arabic Genre Classification

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## Abstract

This paper describes the Northwestern University in Qatar submission to the AraGenre 2026 Shared Task on hierarchical Arabic genre classification. We formulate the task as a definition-guided dense retrieval problem by encoding Arabic documents and English genre definitions into a shared semantic space using the multilingual BGE-M3 model. Genre labels are assigned by retrieving the most semantically similar definition without task specific fine tuning. On the development set, the proposed approach achieved a Broad Macro F1 of 0.6125, a Specific Macro F1 of 0.6125, and a Hierarchical Macro F1 of 0.6125. These results establish a lightweight, training-free baseline while highlighting limitations in generalization, particularly for fine-grained genre classification.

## 1 Introduction

Genre classification is a fundamental task in Natural Language Processing (NLP) that aims to identify the communicative purpose and writing style of a document. It plays an important role in downstream applications such as information retrieval, document organization, recommendation systems, and corpus analysis. Despite the success of pretrained language models, hierarchical genre classification remains challenging due to the semantic overlap between related genres and the need to distinguish documents at multiple levels of granularity. The AraGenre 2026 Shared Task (El-Haj et al., 2026) addresses this challenge by introducing a benchmark for hierarchical Arabic genre classification, where each Arabic document is assigned both a broad genre and a fine-grained genre label .The task requires systems to align Arabic documents with English genre definitions while preserving consistency across the hierarchical taxonomy. To address this challenge, we propose a definition-guided dense retrieval framework based on the multilingual BGE-M3 embedding model. Our approach encodes Arabic input texts and English genre definitions into a shared semantic space and predicts hierarchical labels by retrieving the most semantically similar genre definition without task-specific fine-tuning. On the development set, the proposed framework achieved a Hierarchical Macro F1 score of 0.6125. In the official hidden test evaluation, the system achieved a Hierarchical Macro F1 score of 0.3302 and ranked 15th on the official leaderboard, highlighting the limitations of this lightweight retrieval-based approach on the hidden test distribution.

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## 2 Background

The AraGenre 2026 Shared Task addresses hierarchical Arabic genre classification, where each input consists of an Arabic document and the system predicts both a broad genre and its corresponding specific genre. For example, a document discussing Islamic teachings is assigned the broad genre *Religious* and the specific genre *Religious Commentary*.

The benchmark covers Modern Standard Arabic, Classical Arabic, and multiple Arabic dialects across diverse communicative settings. Participants are provided with English genre definitions together with labeled training and development data, while system evaluation is performed on a hidden test set. The development set contains 110 Arabic documents distributed across six broad genres and six corresponding specific genres. Systems are evaluated using Broad Accuracy, Broad Macro F1, Specific Accuracy, Specific Macro F1, and Hierarchical Macro F1, with Hierarchical Macro F1 serving as the official ranking metric.

Unlike conventional supervised text classification methods, our approach formulates the task as a



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definition-guided dense retrieval problem. Arabic documents and English genre definitions are encoded into a shared semantic embedding space using BGE-M3, and the most similar definition is retrieved using cosine similarity to assign both broad and specific genre labels without task-specific fine-tuning.

Recent multilingual embedding models have demonstrated strong semantic representation and cross-lingual retrieval capabilities by mapping texts from different languages into a shared embedding space (Chen et al., 2024). Dense retrieval approaches have also shown strong effectiveness for semantic retrieval, outperforming traditional sparse lexical methods such as BM25 on open-domain question answering benchmarks (Karpukhin et al., 2020). However, the application of definition-guided dense retrieval to hierarchical Arabic genre classification has received limited attention. Motivated by this observation, we leverage BGE-M3 to retrieve the most semantically similar English genre definition for each Arabic document without task-specific fine-tuning.

## 3 Task Overview

The AraGenre 2026 Shared Task addresses hierarchical Arabic genre classification by requiring systems to predict both broad and specific genre labels. The task provides English genre definitions together with labeled training and development datasets, while evaluation is performed on a hidden Arabic test set. Systems are ranked using Hierarchical Macro F1.

## 4 System Overview

We formulate hierarchical Arabic genre classification as a definition-guided dense retrieval task. Instead of training a supervised classifier, our approach retrieves the most semantically relevant genre definition for each input document. Both the input document and the predefined genre definitions are encoded into a shared semantic embedding space using the multilingual BGE-M3 model. The retrieved definition is then used to assign both the broad and specific genre labels. The proposed system does not use any external training data or task-specific supervision. The only external resource is the pretrained multilingual BGE-M3 embedding model.The proposed retrieval framework directly addresses the cross-lingual nature of the task by matching Arabic documents with English genre definitions in a shared semantic embedding space.

### 4.1 Genre Definition Representation

Each candidate genre is represented by a textual description constructed from the information provided in the shared-task definition files. Specifically, every candidate combines the broad genre label, its textual definition, the specific genre label, and its corresponding definition into a single descriptive text. This representation provides the embedding model with complete hierarchical information for each candidate genre.For each candidate genre, a single textual representation is constructed by concatenating the broad genre label, the corresponding broad genre definition, the specific genre label, and the corresponding specific genre definition in the order provided by the official shared-task definition files. No additional preprocessing or length balancing is applied.

### 4.2 Dense Semantic Encoding

Given an input document $x$ and a candidate genre definition $d_i$, the multilingual BGE-M3 encoder $E(\cdot)$ maps both inputs into dense semantic embeddings:

$$\mathbf{h}_x = E(x),$$

$$\mathbf{h}_{d_i} = E(d_i),$$

where $\mathbf{h}_x$ denotes the dense embedding of the input document and $\mathbf{h}_{d_i}$ denotes the dense embedding of the $i$-th genre definition.

All embeddings are L2-normalized before similarity computation. Since the embeddings are normalized, their inner product is equivalent to cosine similarity.

The implementation uses the default BGE-M3 model provided by the FlagEmbedding library. Both input documents and genre definitions are encoded using dense embeddings with a maximum sequence length of 256 tokens. No instruction prompting, task-specific prompts, external training, or model fine-tuning are applied.

### 4.3 Genre Retrieval

For each input document, the semantic similarity between the document embedding and each candidate genre-definition embedding is computed as

$$s(x, d_i) = \frac{\mathbf{h}_x \cdot \mathbf{h}_{d_i}}{\|\mathbf{h}_x\| \|\mathbf{h}_{d_i}\|},$$

where $s(x, d_i)$ denotes the cosine similarity between the input document $x$ and the $i$-th genre definition $d_i$.

The retrieved genre definition is obtained by selecting the candidate with the highest similarity score:

$$d^* = \arg\max_i s(x, d_i),$$

where $d^*$ denotes the retrieved genre definition.

The broad and specific genre labels associated with the retrieved definition are returned as the final prediction.

### 4.4 Inference Procedure

During inference, genre-definition embeddings are computed once for the candidate definitions, while each input document is encoded independently. A similarity matrix is then constructed by comparing every document embedding with all candidate definition embeddings. For each document, the genre definition with the maximum similarity score is selected as the final prediction. This retrieval-based framework requires no task-specific fine-tuning or classifier training and relies entirely on semantic similarity in the shared embedding space.

## 5 Experimental Setup

The proposed system uses the official AraGenre training (140 instances) and development (110 instances) datasets, while the hidden test set is reserved for the official evaluation. Arabic documents and English genre definitions are encoded using the pretrained multilingual BGE-M3 model (batch size = 16, maximum sequence length = 256). The implementation uses FlagEmbedding, PyTorch 2.11.0+cu128, and scikit-learn 1.9.0. System performance is evaluated using Broad Accuracy, Broad Macro F1, Specific Accuracy, Specific Macro F1, and Hierarchical Macro F1, the official ranking metric.

## 6 Results

The proposed definition-guided dense retrieval framework was evaluated on the AraGenre development set using the official shared-task evaluation metrics. Table 1 summarizes the obtained performance in terms of Broad Accuracy, Broad Macro F1, Specific Accuracy, Specific Macro F1, and Hierarchical Macro F1. Our system ranked 15th on the official hidden test leaderboard.

Table 1: Performance of the proposed system on the AraGenre development set.

| Metric | Score |
|---|---|
| Broad Accuracy | 0.6545 |
| Broad Macro F1 | 0.6125 |
| Specific Accuracy | 0.6545 |
| Specific Macro F1 | 0.6125 |
| Hierarchical Macro F1 | 0.6125 |

Table 2: Official evaluation results on the hidden test set.

| **Metric** | **Score** |
|---|---|
| Broad Accuracy | 0.5494 |
| Broad Macro F1 | 0.4596 |
| Specific Accuracy | 0.2305 |
| Specific Macro F1 | 0.2008 |
| Hierarchical Macro F1 | 0.3302 |

As shown in Table 1, the proposed system achieved a Broad Accuracy of 65.45% and a Specific Accuracy of 65.45%, resulting in a Hierarchical Macro F1 score of 61.25% on the development set. These results indicate that the multilingual BGE-M3 embedding model effectively captures the semantic relationship between Arabic input texts and the corresponding English genre definitions without requiring task-specific fine-tuning.

A closer inspection of the class-wise performance reveals that the system achieved the highest F1 scores for the *Legal* (0.8889) and *Religious* (0.8235) categories, while lower performance was observed for the *Learning* (0.2222) and *Interactive* (0.3902) categories. This variation suggests that certain genres possess more distinctive semantic characteristics, whereas others exhibit greater overlap with related categories, making them more challenging for retrieval-based prediction. Overall, the experimental results demonstrate that the proposed lightweight definition-guided dense retrieval framework is effective for hierarchical Arabic genre classification while maintaining a simple inference pipeline and eliminating the need for supervised classifier training.

### 6.1 Hidden Test Set Results

Table 2 presents the official evaluation results on the hidden test set released by the shared-task organizers. Compared with the development set, performance decreases across all evaluation metrics, reflecting the increased difficulty of the hidden

benchmark. The largest reduction is observed for the Specific Macro F1 score, indicating that fine-grained genre classification remains considerably more challenging than broad genre prediction. Nevertheless, the proposed retrieval-based framework maintains stronger performance at the broad genre level, achieving a Broad Macro F1 of 0.4596 and a Broad Accuracy of 0.5494 on the official test set.

### 6.2 Error Analysis

An analysis of the development-set predictions shows that the retrieval-based approach performs better for semantically distinctive genres than for closely related categories.The most frequent broad-genre confusion occurs between *Informative* and *Interactive* (7 instances), followed by *Interactive* misclassified as *Creative* (7 instances) and *Learning* misclassified as *Informative* (5 instances). In contrast, the *Legal* genre achieves perfect classification, with all 20 development instances correctly predicted.At the specific-genre level, *Diplomatic Communication* achieves perfect classification (20/20), whereas *Educational Explanations* are frequently confused with *Analytical Reports* (5 instances). Similarly, Advice Columns are often predicted as Motivational Writing (7 instances), reflecting the semantic similarity between these communicative functions.Most classification errors occur between a small number of genre pairs, whereas Legal and Diplomatic Communication achieve perfect classification on the development set.

## 7 Conclusion

This paper presented a definition-guided dense retrieval framework for the AraGenre 2026 Shared Task on hierarchical Arabic genre classification. The proposed system uses the multilingual BGE-M3 embedding model to encode Arabic documents and English genre definitions into a shared semantic embedding space and predicts genre labels by retrieving the most semantically similar definition without task-specific fine-tuning.

On the development set, the proposed approach achieved a Hierarchical Macro F1 score of 0.6125. In the official hidden test evaluation, the system ranked 15th on the leaderboard. These results show that multilingual dense retrieval provides an effective and lightweight solution for definition-guided Arabic genre classification.The error analysis indicates that the proposed approach performs well for semantically distinctive genres but remains challenged by closely related genre categories, particularly at the fine-grained classification level.Future work will investigate task-specific fine-tuning and improved semantic representations to enhance fine-grained genre discrimination

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